

Syllabus for CS3923-UG / 6813-GY: Computer Security

New York University

Fall 2026

Professor Evan Johnson

Weekly on Tuesdays 6:30-9PM (Sept 8-Dec 15)

To contact professor: e.johnson@nyu.edu

Office location: 370 Jay Street, room 1055

Office hours: 2:30-3:30 at 370 Jay Street, Room 1055

Graduate teaching assistants: TBD

Course Description

This class provides a firm grounding in computer security concepts and the basics of analyzing and designing secure systems. This course teaches students about threat modeling, software and hardware vulnerabilities (e.g., memory unsafety, side channels, sandbox escapes), and defensive techniques (e.g., exploit mitigations, memory and resource isolation, constant-time programming). Students will work in teams to apply these techniques to attack and harden a simple web-browser.

Course Objectives

During this course, students will:

1. Develop a security mindset while acting ethically, including identifying threats, modeling adversaries, reasoning about trust and attack surfaces, and evaluating security assumptions.
2. Understand fundamental classes of software vulnerabilities such as memory unsafety, side channels, and sandbox escapes as well as how attackers exploit them.
3. Gain practical experience discovering and triaging vulnerabilities through the use of fuzzing, sanitizers, and static analysis.
4. Understand the core mechanisms used to defend computer systems such as memory safety, exploit mitigations, isolation, and constant-time programming.

Course Structure

This is a project-based course organized around five multi-week assignments. Through these assignments, students will work in teams to analyze, attack, and harden a simplified web browser while applying the security concepts and techniques introduced in class. Weekly lectures will provide the conceptual foundations and practical guidance needed to complete each stage of the project.

Readings

There is no textbook for this class. The lectures will often include online materials that need to be read.

Prerequisites:

Prerequisite for Brooklyn Students: CS-UY 2214

Prerequisite for CAS Students: CSCI-UA 201

Prerequisite for Abu Dhabi Students: CS-UH 2010 or ENGR-AD 3511

Prerequisite for Shanghai Students: CENG-SHU 202
Co-requisite for ALL Students: CS-UY 3224

Grading etc:

Exam 1: 20% (exam 1)

Exam 2: 20% (exam 2)

Exam 3: 35% (final)

Projects: 25% (5 assignments total)

Grade Scale

A = 92–100
A– = 90–92
B+ = 88–90
B = 82–88
B– = 80–82
C+ = 78–80
C = 72–78
C– = 70–72
D+ = 68–70
D = 62–68
F = below 62

Course schedule

Week 1 Sept 8	Introduction & Threat Modeling	PA1 released
Week 2 Sept 15	Memory Corruption I: Buffer Overflows	PA1 Due PA2 released
Week 3 Sept 22	Memory Corruption II: Heap Exploits	
Week 4 Sept 29	Memory Corruption III: Finding Vulns	PA2 due PA3 released
Week 5 Oct 6	Memory Safety I: Mitigations	
Week 6 Oct 13	Exam 1	
Week 7 Oct 20	Memory Safety II: Hardware Defenses	PA3 due
Week 8 Nov 3	Memory Safety III: Language Defenses	PA 4 released

Week 9 Nov 10	Isolation I: Processes & Hypervisors	
Week 10 Nov 17	Isolation II: Software Fault Isolation	PA4 due PA5 released
Week 11 Nov 24	Exam 2	
Week 12 Dec 1	Side Channels	
Week 13 Dec 8	Constant-time Programming & IFC	PA 5 due
Week 14 Dec 15	Reading Day	
Week 15 Dec 22	Final	

Religious Accommodation

NYU's [Policy on Academic Accommodations for Religious Holidays and Observances](#) states that students may, without penalty, excuse themselves from academic obligations and otherwise receive reasonable accommodation when required for religious and spiritual holidays and observances. You must notify me in advance of religious holidays or observances that might coincide with exams, assignments, or class times to schedule reasonable alternatives. Students may also contact religiousaccommodations@nyu.edu for assistance.

Academic Dishonesty

Students must follow the collaboration guidelines for the projects. All tests and quizzes are individual. Students must behave ethically at all times. As an NYU community member, it is your responsibility to know your rights and responsibilities around academic misconduct. Please [click here](#) to review the NYU Tandon Student Code of Conduct. Any questions about how this policy relates to this class, please ask your professor.

Generative AI Tools

You may use ChatGPT or a similar tool for the assignments of this class, but you must document this in code comments, assignment sheets, and similar. You will need to understand security concepts in a deep manner for the exams (where Generative AI Tools are not permitted), so I suggest you use these tools carefully and sparingly.

The Office of Student Advocacy

At NYU Tandon, we recognize that medical, mental health, or personal issues can sometimes affect your attendance and academic performance. If you are experiencing challenges that you feel are impacting your ability to attend class or succeed in your coursework, we encourage you to reach out to the NYU

Tandon Office of Student Advocacy. The Student Advocacy team is here to connect you with the resources and support you need to address these challenges and ensure you can continue to thrive academically. If you begin experiencing any issues that may affect your attendance or academic performance, please reach out to the NYU Tandon Office of Student Advocacy as soon as possible as we will be unable to appropriately address these concerns as the semester progresses, so early communication is essential to ensure the support and resources you need. The NYU Tandon Office of Student Advocacy can be reached at advocacy.tandonstudentlife@nyu.edu and is located in 5 MetroTech Center LC240.

Inclusion

NYU is committed to creating an inclusive and equitable environment for all students. In this class, I aim to cultivate a strong sense of community, emphasizing that it is a space where individuals of diverse backgrounds—spanning beliefs, ethnicities, national origins, gender identities, sexual orientations, religious and political affiliations, and abilities—are treated with respect. All members of the class are expected to contribute to an inclusive atmosphere for each other. While disagreement and differing ideas are welcome, the expectation is that we respect perspectives beyond our own, even when they diverge from our personal beliefs or experiences. If you encounter disrespect or discrimination in the class or feel that the outlined standards are not being upheld, please report your experiences to me.

Moses Center Statement of Disability

If you are a student with a disability who is requesting accommodations, please contact New York University's Moses Center for Students with Disabilities (CSD) at 212-998-4980 or mosescsd@nyu.edu. You must be registered with CSD to receive accommodations. Information about the Moses Center can be found at www.nyu.edu/csd. The Moses Center is located at 726 Broadway on the 3rd floor.

NYU School of Engineering Policies and Procedures on Academic Misconduct – complete Student Code of Conduct [here](#)

A. Introduction: The School of Engineering encourages academic excellence in an environment that promotes honesty, integrity, and fairness, and students at the School of Engineering are expected to exhibit those qualities in their academic work. It is through the process of submitting their own work and receiving honest feedback on that work that students may progress academically. Any act of academic dishonesty is seen as an attack upon the School and will not be tolerated. Furthermore, those who breach the School's rules on academic integrity will be sanctioned under this Policy. Students are responsible for familiarizing themselves with the School's Policy on Academic Misconduct.

B. Definition: Academic dishonesty may include misrepresentation, deception, dishonesty, or any act of falsification committed by a student to influence a grade or other academic evaluation. Academic dishonesty also includes intentionally damaging the academic work of others or assisting other students in acts of dishonesty. Common examples of academically dishonest behavior include, but are not limited to, the following:

1. Cheating: intentionally using or attempting to use unauthorized notes, books, electronic media, or electronic communications in an exam; talking with fellow students or looking at another person's work during an exam; submitting work prepared in advance for an in-class examination; having someone take an exam for you or taking an exam for someone else; violating other rules governing the administration of examinations.
2. Fabrication: including but not limited to, falsifying experimental data and/or citations.
3. Plagiarism: intentionally or knowingly representing the words or ideas of another as one's own in any academic exercise; failure to attribute direct quotations, paraphrases, or

borrowed facts or information.

4. Unauthorized collaboration: working together on work meant to be done individually.

5. Duplicating work: presenting for grading the same work for more than one project or in more than one class, unless express and prior permission has been received from the course instructor(s) or research adviser involved.

6. Forgery: altering any academic document, including, but not limited to, academic records, admissions materials, or medical excuses.

[Student absence policy](#)

[Absence reporting form](#)